Language-Guided Object Localization via Refined Spotting Enhancement in Remote Sensing Imagery

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***Abstract*— Language-guided remote sensing image object localization uses intuitive natural language interactions to locate objects of interest within satellite or drone imagery, and has a wide range of practical applications. Early research on this task typically used discriminative models that relied on predefined task heads, limited to locating a single object and lacking flexibility. In recent years, multimodal large language models (MLLMs)-based generative models leverage their language understanding ability to comprehend more complex language references and provide flexible outputs. However, these models tend to be too large and have limited accuracy in spotting dense, small objects in remote sensing scenarios. The reasons are: 1) generative models treat continuous coordinate prediction as a token classification problem, which fails to reflect the actual localization gap and 2) current methods typically use CLIP pre-trained encoders that align text-image pairs at a global level, and may not capture the fine-grained semantic object information necessary for accurate localization. To address these shortcomings, we introduce a lightweight generative model named the localization model with refined spotting enhancement (LM-RSE). Refined spotting enhancement includes the refinement of bounding box (BBox) outputs and the integration of fine-grained semantic features. Specifically, we design a BBox refinement (BBR) approach that includes a special token, a BBox decoder, and a custom regression loss function to refine the spotting precision and optimize the training process. Additionally, we propose a fine-grained semantic integration (FGSI) strategy, integrates a fine-grained image encoder, a vision-language semantic processing (VLSP) layer, and a two-stage, full-parameter training strategy, all working together to effectively enhance the granularity of spotting. Building on this, we further refine the language-guided object localization task into two types: expression-guided and class-guided. For the former, we utilize the RSVGD dataset for evaluation and achieved state-of-the-art performance; for the latter, our evaluation results surpassed those of GeoChat. Our code and checkpoints will be released at https://github.com/Zhang-Peirong/LM-RSE.**

***Index Terms*— Bounding box refinement (BBR), fine-grained semantic features, language-guided object localization, vision-language models.**

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I. Introduction

LANGUAGE-GUIDED object localization [1] in remote sensing imagery is an emerging task that involves the use of natural language to spot and locate objects within satellite or aerial imagery. The emergence of this technology provides a user-friendly approach for non-expert users to easily retrieve and access object information in remote sensing images, making remote sensing technology more accessible and efficient. The practical applications of this task are extensive and varied, including critical areas such as military object identification, disaster response coordination, agricultural monitoring, and environmental surveillance. In spite of its importance, this task is still a relatively emerging field that has not been fully explored.

As shown in Fig. 1, unlike traditional object detection [2], [3], [4] tasks that can only detect all objects of fixed categories in an image, language-guided object localization offers higher interactivity and flexibility. Objects in remote sensing images are typically diverse and densely arranged, and not all objects are equally important or relevant for localization tasks. In specific application scenarios, we may only be interested in objects of certain categories, specific locations, or those with particular characteristics (such as color, size, etc.) [5], [6]. This is where language-guided object localization becomes particularly valuable. It enables the spotting of specific objects within a remote sensing image that precisely match the provided language description, which can effectively capture user interests and guide the model's operation.

Consequently, language-guided object localization emerged from the earlier concept of visual grounding [5], [6], [7], reflecting its evolution in response to the need for more intuitive and user-driven interactions in model training and deployment. Sun et al. [6] first introduced the concept of remote sensing visual grounding and proposed the GeoVG method, while Zhan et al. [5] developed the MGVLF approach. These methods are both discriminative models. They employed separate visual and language encoders to process image and language inputs, respectively, followed by a fusion module

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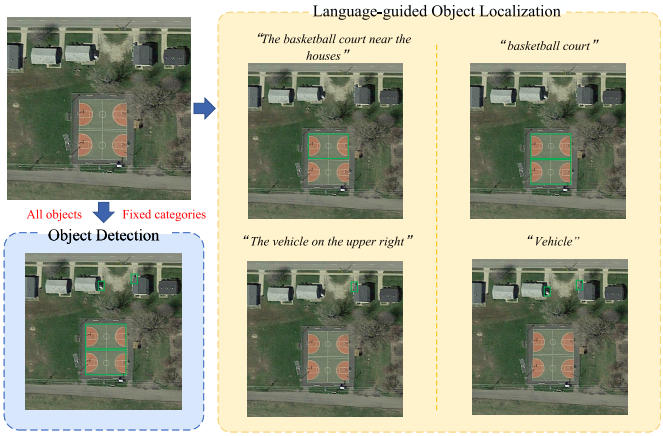


Fig. 1. Comparison between traditional object detection tasks and language-guided localization tasks. Traditional object detection can only detect all objects of fixed categories in the image. Language-guided localization tasks can locate objects of interest in the image based on language (expression descriptions or category information), offering higher interactivity and flexibility.

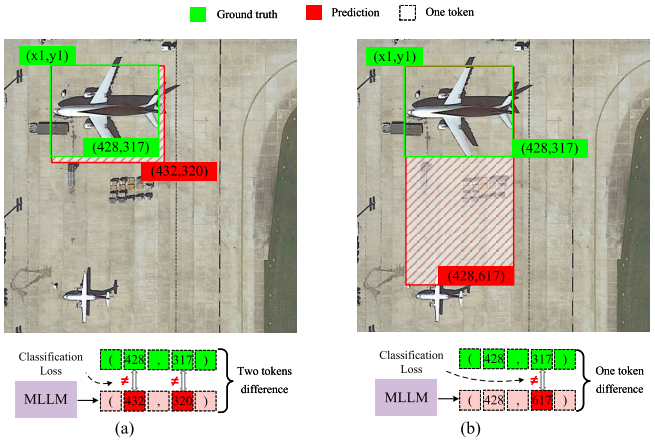


Fig. 2. MLLMs treat numerical regression as a classification task. (a) and (b) Illustrate the difference between the IOU distance and the loss used by LLMs. In (a), prediction and the ground truth have a smaller IoU but incur a larger loss for LLMs. In contrast, (b) shows a situation where the IoU is quite large, yet the loss is smaller.

to align these features and task-specific heads for object localization. However, these heads and the corresponding training methods were primarily designed for single-object localization [1], which means they were optimized to focus on spotting one object that best matched the given description. This design choice, coupled with the limited availability of diverse training data [8], restricted the models' ability to generalize to more complex scenarios. As a result of these limitations, the inherent architecture of these discriminative models lacked the flexibility and scalability required for more advanced applications.

Recognizing these limitations, researchers have begun exploring alternative approaches. With the breakthrough progress in multimodal large language models (MLLMs) [9], there have been increasing attempts [10], [11], [12], to apply them in the field of language-guided object localization. MLLMs, built on large language models (LLMs) [13], [14] trained with extensive corpora, can leverage their language understanding ability to comprehend more complex language references and provide flexible outputs, offering greater interactivity. Unlike traditional discriminative models, generative models such as MLLMs offer significant advantages in handling multiobject localization tasks. By framing the prediction of bounding boxes (BBoxs) as a generative task, these models can more effectively manage the complexity of spotting and localizing multiple objects within an image. To enable MLLMs to predict object BBoxs, Peng et al. [12] used MAGNETO as a language decoder, integrating discretized images into 32×32 bins that were added to the language model's vocabulary for predicting object coordinates. Shikra [11], on the other hand, reused the original vocabulary and transformed the BBox coordinates into normalized values with certain precision. These models have shown promise in handling unique object localization, as well as multiobject localization tasks. Using MLLMs for localization tasks has become a new research trend.

However, these models are often excessively large, with sizes reaching up to 7 or 13 billion parameters, making them expensive to train and difficult to deploy [15], [16]. More importantly, remote sensing imagery often contains tiny objects and densely arranged objects [17], making it difficult for these MLLM-based models to achieve precise localization. Specifically, this difficulty arises due to two main reasons: 1) MLLMs treating numerical regression as a classification task and 2) inadequate extraction of fine-grained semantic features.

As generative models, MLLMs typically treat the prediction of object coordinates as a classification problem rather than a regression task involving continuous variables [18]. As shown in Fig. 2, compared to (b), scenario (a) exhibits a bigger intersection over union (IoU) between the predicted BBox and the ground truth. However, for MLLMs employing cross-entropy as the loss function, the discrepancy in loss between (b) and the ground truth is smaller than that observed in (a). This methodology neglects the numerical proximity of values, as well as the spatial relationships between coordinate points, thereby presenting an inherent limitation that considerably undermines localization accuracy.

Furthermore, in the context of language-guided object localization, it is crucial to extract fine-grained semantic features understandable by the language models from remote sensing images [19]. As shown in Fig. 3(a), current methods often employ CLIP pre-trained image encoders [20] for visual feature extraction. These methods typically use contrastive learning to align entire images with their corresponding captions, which limits their ability to capture dense semantic information within the image. This inadequate extraction of fine-grained semantic features struggles to accurately model and respond to dense object instances within remote sensing imagery, thereby affecting the overall performance of localization tasks.

To address those shortcomings and develop a user-friendly remote sensing localization model, we propose localization model with refined spotting enhancement (LM-RSE), an innovative, lightweight framework designed specifically for the remote sensing language-guided object localization task.

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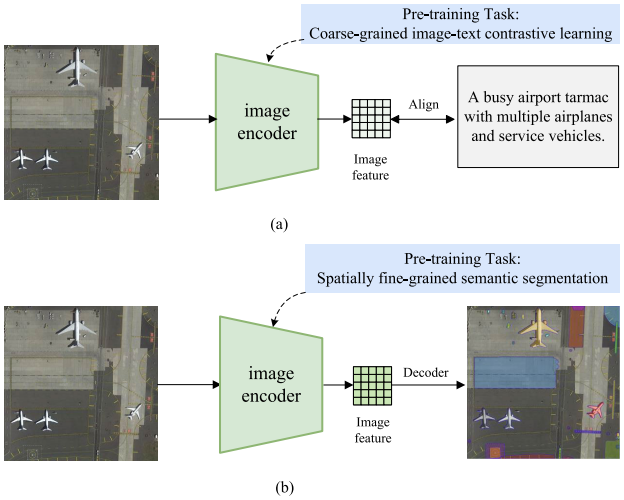


Fig. 3. Comparison of pre-training tasks between CLIP and SAM. (a) Previous methods often employ CLIP pre-training encoders, using contrastive learning to align images with text. This approach leads to inadequate extraction of fine-grained semantic features. (b) Pre-training task of SAM, which is trained for semantic segmentation and could extract fine-grained semantic features. Our image encoder leverages the strengths of both and is capable of extracting fine-grained, semantically rich features.

LM-RSE is an interactive generative model that integrates a fine-grained image encoder with a compact language model as its decoder. Refined spotting enhancement includes the refinement of BBox outputs and the integration of fine-grained semantic features. Specifically, a BBox refinement (BBR) approach and fine-grained semantic integration (FGSI) strategy are involved to significantly enhance the accuracy of language-guided object localization in remote sensing scenarios.

1) *BBR approach:* To refine the spotting precision, we designed the BBR approach, which includes a special token, a BBox decoder, and a novel loss function. This approach introduces a dedicated branch that leverages the hidden states of the specific token `<BBox>` and utilizes the BBox decoder to generate regressed object coordinates. Additionally, a custom regression loss function is implemented to supervise and optimize the training process. By addressing the inherent weaknesses in treating numerical coordinates as categorical data, our approach ensures that the distance between prediction and ground truth is minimized, thus significantly improving the precision of localization tasks.
2) *FGSI strategy:* We design an FGSI strategy to integrate fine-grained semantic features and preserve semantic integrity during training. Structurally, it includes a fine-grained image encoder to extract detailed fine-grained semantic visual features, and a vision-language semantic processing (VLSP) layer to efficiently integrate these features into the language model's dimensional space. Additionally, a two-stage, full-parameter training strategy is designed to preserve the richness of semantic information. The fine-grained image encoder is initialized with SAM [21]'s pre-trained encoder weights and acquires fine-grained semantic information through joint training with the VLSP layer and the language model. This strategy enhances the model's ability to extract fine-grained semantic features and ensures these features are effectively leveraged during training to guide the language model in achieving high-precision localization.

We also categorize the language-guided object localization task in remote sensing into two types: expression-guided and class-guided object localization. In expression-guided object localization, the goal is to locate a unique object in the image based on descriptive text, or "expression," which specifies attributes like category, color, size, and location. In class-guided object localization, the objective is to identify all objects of a specific category within the image. Unlike traditional object detection, which passively detects all object categories, class-guided object localization actively focuses on identifying objects of a specified class. Building on this categorization, LM-RSE offers a comprehensive solution that effectively handles both expression-guided and class-guided object localization in remote sensing. By integrating a fine-grained image encoder, a compact language model, and leveraging the refined spotting enhancement—which includes both the BBR approach and the FGSI strategy—our model ensures precise and targeted object localization based on specific language inputs.

In summary, the main contributions of our work can be summarized as follows.

1) We introduce an interactive, lightweight generative model named LM-RSE for remote sensing language-guided object localization. This model can process high-resolution remote sensing images, handle both expression-guided and class-guided localization, and offer user-friendly interaction.
2) We propose a novel BBR approach to enhance numerical prediction accuracy in localization tasks, and a new loss function to accelerate convergence and improve precision.
3) We design an FGSI strategy, leveraging a fine-grained image encoder with a VLSP layer and a two-stage, full-parameter training strategy to preserve semantic integrity.
4) LM-RSE achieves state-of-the-art performance on the RSVGD dataset for expression-guided localization and excels in high-precision metrics. For class-guided localization, our model outperforms GeoChat and demonstrates multiobject localization capabilities.

The rest of this article is organized is follows. In Section II, we briefly summarize the related work relevant to our method. Section III introduced the design details of our proposed model LM-RSE. In Section IV, we demonstrate the experimental results of LM-RSE and give a comparison between our model and other baseline models. Finally, we conclude our work in Section V.

II. Related Works

In this section, we provide a concise review of the foundational research relevant to our study. Specifically, it included

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discriminative language-guided object localization and generative language-guided object localization.

A. Discriminative Language-Guided Object Localization

Language-guided localization, also known as referring expression comprehension [22], [23] or visual grounding [24], is a task that combines computer vision and natural language processing (NLP). The traditional approach to this task employs discriminative models, which are generally divided into two-stage, one-stage, and transformer-based methods [25].

Two-stage methods [26], [27], [28], [29] typically involve first detecting objects and then matching the query expression with the candidate objects, with their performance heavily relying on the accuracy of the detector [30]. In contrast, one-stage methods [30], [31] do not involve an explicit region proposal process and are typically based on dense anchors. Although these methods generally offer faster speeds, they require more complex models.

Recently, Transformer-based methods [32], [33] have gained significant popularity. Visual-language transformers typically consist of a visual encoder, a text encoder, and a multimodal fusion module [34]. Li and Sigal [35] utilize a transformer architecture to design a simple one-stage multitask framework for both REC and RES tasks. Deng et al. [7] present the TransVG framework, which includes a stack of transformer encoder layers for multimodal fusion and reasoning, direct coordinates regression for visual grounding, and a visual-linguistic transformer for cross-modal relation reasoning. Yang et al. [36] enhance the performance of the visual grounding task by establishing text-conditioned discriminative features and conducting multistage cross-modal reasoning.

In remote sensing, Zhan et al. [5] introduced the RSVGD benchmark dataset and a novel Transformer-based multilevel cross-modal (MLCM) module to address scale variation and complex backgrounds. Sun et al. [6] focused on complex geospatial relationships and large-scale scenes in remote sensing images. However, these methods lack interactivity and flexibility, and can only locate a single language-guided object.

B. Generative Language-Guided Object Localization

Generative methods using LLMs [13], [14] offer significant improvements in interaction and flexibility for solving localization problems. These approaches involve integrating boxes or regions of an image into MLLMs, thereby enhancing the adaptability and performance of localization tasks.

Zhang et al. [37] introduce GPT4ROI, which uses spatial instruction tuning by incorporating BBox references into language instructions for accurate region referring. Chen et al. [38] present Pix2seq, which represents object BBox coordinates as text tokens, allowing the language model to output object locations via token classification. Yang et al. [39] extend Pix2seq's idea to vision and language tasks such as visual grounding with the proposed UniTAB, using a transformer encoder–decoder with a unified decoding sequence including `<obj>` tokens for text and box outputs. Building on this, Peng et al. [12] utilize Kosmos-2, applying the token classification concept to MLLMs. Kosmos-2 discretizes images into 32×32 bins for coordinate prediction. These positional points are incorporated into the language model's vocabulary, enabling the model to predict object coordinates. Chen et al. [11] introduce Shikra, which reuses the original vocabulary and transforms the BBox coordinates into normalized values with certain precision. Zhang et al. [40] propose NExT-Chat, which introduces a `<trigger>` token for decoding the BBoxs and segmentation masks, further enhancing the efficiency and effectiveness of generative language-guided object localization methods.

In remote sensing, Kuckreja et al. [10] introduce GeoChat, the first LLMs-based vision-language model for remote sensing. They combine a CLIP-ViT(L-14) [20] as the visual backbone with Vicuna-v1.5(7B) serving as the foundation LLM, enabling multitask conversational capabilities. However, these models are large, with high deployment and training costs.

III. Methods

In this section, we introduce the details of our proposed model, LM-RSE. It contained four key areas: overview, data preparation, BBR approach, and FGSI strategy. We first overview the architecture of LM-RSE in Section III-A. Then, we introduce the data preparation process of LM-RSE in Section III-B. Sections III-C and III-D delineate the design of the BBR approach and FGSI strategy.

A. Overview

LM-RSE is an encoder–decoder structure, lightweight, and interactive generative model capable of handling both expression-guided and class-guided localization. As illustrated in Fig. 4, the core components of the model are a fine-grained image encoder and a compact language decoder. A visual-language semantic projection layer (VLSP layer) is designed to map fine-grained semantic visual features into the language model's dimension while preserving their semantic integrity. The BBR branch is incorporated after the language decoder as to enhance localization precision.

To effectively handle both expression-guided and class-guided localization within a single model, we utilize an encoder–decoder architecture, which allows for flexible integration of visual and language information, enabling the model to accurately process and generate outputs tailored to both unique object identification and class-based localization tasks. In this architecture, remote sensing images are first processed by the visual encoder to extract visual features. These features are then passed through the VLSP layer, which bridges the encoder and the decoder. The output from the VLSP layer is combined with text embeddings derived from the input prompts to form multimodal embeddings, which are then fed into a compact language decoder, which is OPT-125M.

To enhance the precision of localization, an additional BBR branch is incorporated after the language decoder. This branch refines the numerical predictions related to localization tasks, thereby improving the model's overall accuracy. Additionally, the BBR approach introduces an extra loss term during training, which helps fine-tune the model's performance.

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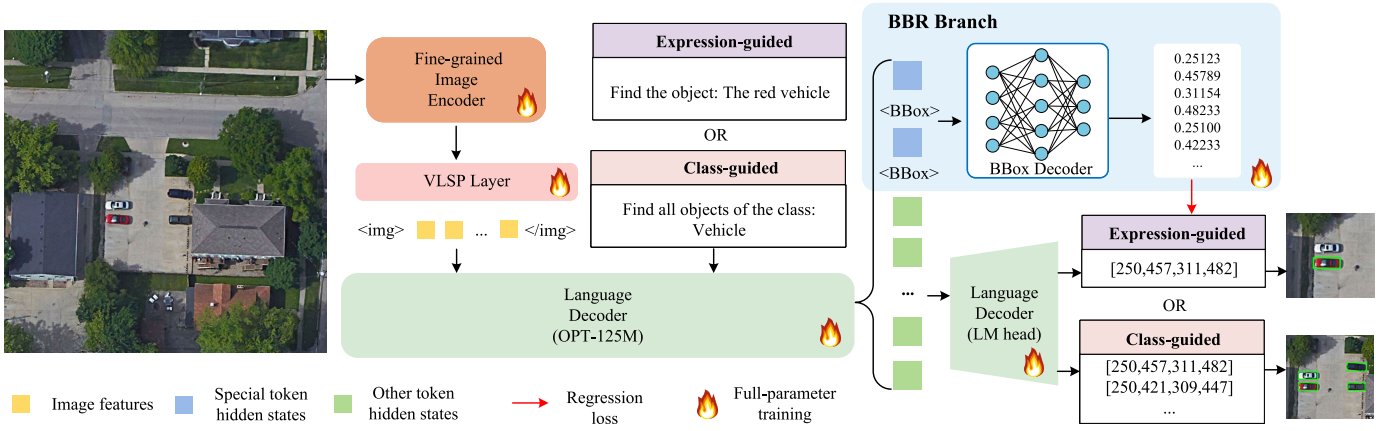


Fig. 4. Overall architecture of LM-RSE. We integrate a fine-grained image encoder and a VLSP layer to get fine-grained semantic feature, which are then decoded by a language decoder (OPT-125M) in either expression-guided or class-guided prompt. The special token <BBox> and BBR branch refine BBoxs for precise localization by providing an auxiliary regression loss for training.

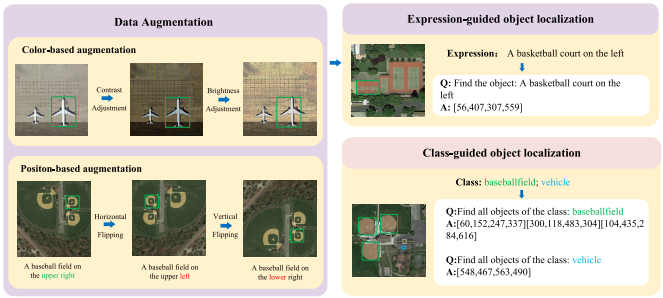


Fig. 5. Data preparation process. For expression-guided localization, the color-based and position-based augmentations are first applied, then the visual grounding dataset are formatted with task-specific prompt in a dialog format. For class-guided localization, object detection datasets are transformed based on their categories.

The architecture design and training stages emphasize the FGSI strategy. We employ a fine-grained image encoder initialized with SAM's encoder and introduce a simple and lightweight VLSP layer, alongside a two-stage, full-parameter training process. This approach effectively extracts fine-grained semantic features and achieves lossless integration.

Overall, the LM-RSE model's architecture is designed to integrate fine-grained semantic features effectively and enhance the precision of language-guided object localization tasks. By leveraging the powerful interaction capabilities of language models, the novel BBR approach, and the FGSI strategy, our model addresses both expression-guided and class-guided localization in remote sensing with improved accuracy and efficiency.

B. Data Preparation

To leverage the language understanding capabilities of the language model and stimulate the model's ability for expression-guided and class-guided localization, as shown in Fig. 5, we reformat the remote sensing visual grounding and object detection datasets into a dialog format. This unification process involves converting the data into a structure that facilitates interaction, enabling the LM-RSE to better handle different localization tasks.

1) Data Augmentation: To address the scarcity of visual grounding datasets for expression-guided object localization in remote sensing, we expanded the dataset using a semantic-preserving data augmentation pipeline. The goal was to increase dataset diversity while maintaining the alignment between textual descriptions and target objects. The pipeline includes two main types of augmentations: color-based (contrast and brightness adjustments) and position-based (horizontal/vertical flipping, minor angle rotations, and translations). A key component of the position-based augmentation is the text-aware flipping technique, which preserves the spatial relationships between objects during flips. For example, when an object like an airplane is on one side of the image (e.g., left), it moves to the opposite side (e.g., right) after flipping. To preserve semantic meaning, the pipeline detects directional terms (e.g., "left," "right," "top," and "bottom") in the textual descriptions. If present, the pipeline dynamically adjusts the positional information in the text to reflect the new object positions. This ensures that textual annotations stay aligned with the augmented image, preserving both visual and semantic integrity.

2) Expression-Guided Object Localization: The input for expression-guided localization task is a pair consisting of an image and a textual description. Mathematically, it can be represented as $X = (\text{Img}, \text{expression})$, where Img is the image data and expression is the corresponding text data. The output for the task is typically a set of coordinates of a BBox within the image Img that corresponds to region described by query. This can be represented as $Y = [x_1, y_1, x_2, y_2]$, where (x_1, y_1) and (x_2, y_2) are the coordinates of the top-left and bottom-right corners of the BBox, respectively. An example dialog template of expression-guided localization for our model are defined as follows.

User: "⟨img⟩ Find the object: expression"
Model: "$[x_1, y_1, x_2, y_2]$"

3) Class-Guided Object Localization: In a class-guided localization task, the input includes both textual and visual modalities. The inputs are represented as $X = (\text{Img}, \text{Class})$, where Img denotes the image data and Class denotes the

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category of the object to be detected in textual format. The outputs are represented as $Y = \{\text{BBox}_1, \text{BBox}_2, \ldots, \text{BBox}_n\}$, where n is the number of objects in the image Img belonging to the specified category, varying across different images. The ith BBox is represented as $[x_1^i, y_1^i, x_2^i, y_2^i]$, where (x_1^i, y_1^i) are the coordinates of the top-left corner and (x_2^i, y_2^i) are the coordinates of the bottom-right corner. An example dialog template for class-guided localization in our model is designed to facilitate this process:

User: "⟨img⟩ Find all objects of the class: class."
Model: "$[x_1^1, y_1^1, x_2^1, y_2^1]$, $[x_1^2, y_1^2, x_2^2, y_2^2], \ldots,$ $[x_1^n, y_1^n, x_2^n, y_2^n]$*"*

For expression-guided and class-guided localization tasks, the position of the token `<img>` is then replaced by the image feature from the VLSP layer. The dialog format data is subsequently processed by the language decoder using causal masked language modeling during training. However, relying solely on the language model for object localization results in poor precision. To address this, we introduce the BBR approach and the FGSI strategy to enhance the performance of both expression-guided and class-guided localization.

C. BBR Approach

1) BBox Decoder: The BBR approach is designed to improve the handling of BBox coordinates by introducing the special token `<BBox>` into the model's outputs. When the model predicts this token, it generates a hidden state embedding $x_{\text{bbox}} \in \mathbb{R}^{d_{\text{opt}}}$, which is specifically associated with `<BBox>`, where d_{opt} is the dimension of the language decoder. This embedding then triggers additional predictions and refinements for BBox coordinates within the BBox decoder.

Initially, the embedding x_{num} is processed by a three-layer multilayer perceptron (MLP). The first and the second layers transform the input embedding as follows:

$$h_1 = \text{ReLU}(W_1 x_{\text{num}} + b_1) \tag{1}$$

$$h_2 = \text{ReLU}(W_2 h_1 + b_2) \tag{2}$$

where $W_1 \in \mathbb{R}^{d_1 \times (d_{\text{opt}} \times N_{\text{token}})}$, $W_2 \in \mathbb{R}^{d_2 \times d_1}$ and $b_1 \in \mathbb{R}^{d_1}$, $b_2 \in \mathbb{R}^{d_2}$ are the weight matrix and bias vector for the first and second layers, respectively. The number of special tokens is denoted as N_{token}, while d_1 and d_2 represent the hidden dimensions of the MLP. The ReLU activation function introduces nonlinearity, enabling the network to learn complex patterns.

The final layer of the MLP generates the output

$$h_3 = W_3 h_2 + b_3 \tag{3}$$

where $W_3 \in \mathbb{R}^{d_{\text{num}} \times d_2}$ and $b_3 \in \mathbb{R}^{d_{\text{num}}}$ are the weight matrix and bias vector of the third layer. This linear transformation produces the intermediate representation $h_3 \in \mathbb{R}^{d_{\text{num}}}$. The parameter d_{num} represents the maximum number of numerical values that can be refined within a single image.

To ensure the predicted BBox coordinates are within the normalized range $[0, 1]$, the output h_3 is passed through a sigmoid activation function

$$\text{BBox Decoder}(x_{\text{num}}) = \sigma(h_3) \tag{4}$$

where $\sigma(z) = (1/(1 + e^{-z}))$ is the sigmoid function applied element-wise to the vector h_3. This normalization step guarantees that each element of the output vector lies within a valid range. This design maintains the model's stability and ease of use while improving the accuracy of numerical predictions.

2) Loss Function: The total loss $\mathcal{L}_{\text{total}}$ for our model is formulated as the sum of the language loss $\mathcal{L}_{\text{lang}}$ and the regression loss $\mathcal{L}_{\text{regression}}$. This combination ensures balanced optimization between language generation and numerical accuracy. Specifically, the total loss is given by

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{lang}} + \underbrace{\mathcal{L}_{\text{center}} + \mathcal{L}_{\text{bbox}}}_{\mathcal{L}_{\text{regression}}}. \tag{5}$$

Here, the regression loss $\mathcal{L}_{\text{regression}}$ is composed of two key components: the center point loss $\mathcal{L}_{\text{center}}$ and the BBox loss $\mathcal{L}_{\text{bbox}}$. The center point loss $\mathcal{L}_{\text{center}}$ specifically penalizes deviations between the predicted and ground-truth center points of the BBoxs, promoting accurate localization of the object. On the other hand, the BBox loss $\mathcal{L}_{\text{bbox}}$ measures the accuracy of the predicted BBox coordinates relative to the ground truth, ensuring that the model precisely captures the size and position of the objects. Together, these two components work in tandem to refine the model's ability to generate precise and well-aligned BBoxs around the target objects.

a) Language loss: OPT model uses causal masked language modeling and employs the cross-entropy loss function as the language loss to measure the discrepancy between the predicted token probabilities and the true token labels. Since the model predicts each token sequentially based on the previous tokens, it is essential to ensure that the logits and true labels are correctly aligned for accurate loss calculation. Specifically, the logits tensor $L \in \mathbb{R}^{(T+1)\times|V|}$, where T is the sequence length and $|V|$ is the vocabulary size, represents the predicted probabilities for the next token at each step. The additional position accounts for the end-of-sequence token. To align these predictions with the actual subsequent tokens, the logits tensor L is shifted forward by one position. Similarly, the true labels tensor $Y \in \mathbb{R}^{(T+1)}$ is also shifted to maintain alignment with the logits. This shift ensures that the logits predicted by the model at step t are aligned with the corresponding actual labels at the $(t + 1)$th step. After these shifts, both tensors have a sequence length of T.

Before computing the language loss, each time step of the shifted logits $L' \in \mathbb{R}^{T\times|V|}$ first applies a softmax function to produce a probability distribution over the vocabulary

$$\hat{L}_{t,i} = \frac{\exp\left(L'_{t,i}\right)}{\sum_{j=1}^{|V|} \exp\left(L'_{t,j}\right)} \tag{6}$$

where $\hat{L}_t \in \mathbb{R}^{|V|}$ represents the probability distribution at time step t over the entire vocabulary after the softmax transformation, and $\hat{L}_{t,i}$ is the probability of the ith word in the vocabulary at time step t.

The cross-entropy loss is then computed between the shifted logits and labels, defined as

$$\mathcal{L}_{\text{lang}}(\theta) = -\mathbb{E}_{(y,v)\sim\mathcal{D}}\left[\frac{1}{T}\sum_{t=1}^{T} \log P_\theta(\hat{L}_t = y_t \mid y_{<t}, v)\right] \tag{7}$$

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where v denotes the image feature from the VLSP layer, $\hat{L} \in \mathbb{R}^{T\times|V|}$ represents the softmax output of the shifted logits, $y \in \mathbb{R}^T$ represents the shifted labels, and $\mathcal{D}$ denotes the datasets. The term $P_\theta(\hat{L}_t = y_t \mid y_{<t}, v)$ represents the predicted probability for the true label y_t at time step t, given the preceding sequence $y_{<t}$ and the image feature v. The expectation $\mathbb{E}_{(y,v)\sim\mathcal{D}}$ indicates that the loss is averaged over all examples in the dataset $\mathcal{D}$. This loss function effectively penalizes the model for deviations between the predicted probabilities and the actual labels, guiding the model to improve its token predictions over time.

b) Regression loss: To ensure the precision of BBox coordinates in the model's outputs, the special token <BBox> triggers the BBR branch for additional prediction refinement. The BBox decoder processes the hidden states embedding associated with this token to produce more accurate BBox coordinates. To guide the BBox output, a center loss and a bbox loss are incorporated for the regression loss $\mathcal{L}_{\text{regression}}$ during training

$$\mathcal{L}_{\text{regression}} = \mathcal{L}_{\text{bbox}} + \mathcal{L}_{\text{center}}. \tag{8}$$

The center loss and bbox loss work together to accelerate the model's convergence and improve the accuracy of the model's BBox predictions. Given the output of the bbox decoder $w \in \mathbb{R}^{d_{\text{num}}}$, which is then resized to represent the coordinate of the bbox, $\hat{w} \in \mathbb{R}^{4\times N}$. N denotes the number of objects that are referred to by language in the image, which is 1 in expression-guided localization and n_{class} in class-guided localization. It can be defined as

$$N = \begin{cases} 1, & \text{expression-guided localization} \\ n_{\text{class}}, & \text{class-guided localization.} \end{cases} \tag{9}$$

The predicted and ground-truth bboxes are of the format of (x_1, y_1, x_2, y_2). The bbox loss is defined as

$$\mathcal{L}_{\text{bbox}}(\theta) = \mathbb{E}_{u\sim\mathcal{D}}\left[1 - \frac{1}{N}\sum_{n=1}^{N}\text{GIoU}(\hat{w}^n, u^n)\right]_{\text{masked}} \tag{10}$$

where u^n represents the nth ground truth of the image and $\hat{w}^n$ denotes the nth prediction as previously mentioned. The expectation $\mathbb{E}_{u\sim\mathcal{D}}$ indicates that the loss is averaged over all examples in the dataset $\mathcal{D}$.

The Generalized IoU (GIoU) [41] is used to measure the similarity between the predicted BBoxs $\hat{w}^n$ and the ground-truth BBoxs u^n. It is defined as follows:

$$\text{GIoU} = \frac{|\hat{w}^n \cap u^n|}{|\hat{w}^n \cup u^n|} - \frac{|\Omega - (\hat{w}^n \cup u^n)|}{|\Omega|} \tag{11}$$

where $|\hat{w}^n \cap u^n|$ denotes the area of intersection between them, and $|\hat{w}^n \cup u^n|$ represents their combined area. The smallest enclosing box Ω is the minimum box that can entirely contain both, and $|\Omega - (\hat{w}^n \cup u^n)|$ measures the area of Ω that is not covered by the union part.

In addition to using the bbox loss to optimize the IoU between the predicted BBoxs and the ground truth, we also added a center loss to accelerate convergence and improve the robustness of training. The center loss is defined as

$$\mathcal{L}_{\text{center}}(\theta) = \mathbb{E}_{u\sim\mathcal{D}}\left[\delta(\mathcal{C}(\hat{w}), \mathcal{C}(u))\right]_{\text{masked}} \tag{12}$$

$$\mathcal{C}((x_1, y_1, x_2, y_2)) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \tag{13}$$

where u represents the normalized ground-truth BBox coordinates within an image, and $\mathcal{C}$ denotes the function that calculates the center point coordinates from the given corner coordinates. The expectation $\mathbb{E}_{u\sim\mathcal{D}}$ indicates that the loss is averaged over all examples in the dataset $\mathcal{D}$. Since the number of bboxes varies for each sample, each vector of numerical ground-truth values is padded with "nan" values to a fixed length of d_{num} elements, facilitating parallelized training across batches. The term "masked" refers to the non-padded (non-nan) elements of the predictions and ground truth, ensuring that the padding does not interfere with the loss computation. And δ is defined as

$$\delta(x, y) = \begin{cases} 0.5(x - y)^2, & \text{if } |x - y| < \epsilon \\ |x - y| - 0.5\epsilon, & \text{otherwise} \end{cases} \tag{14}$$

where ϵ is a threshold parameter that determines the transition point between the quadratic and linear regions. For differences $|x-y|$ less than ϵ, the loss is quadratic, providing smoothness. For larger differences, the loss is linear, reducing the impact of outliers.

This comprehensive approach ensures that both language generation and BBox prediction are effectively optimized, leveraging the strengths of each component to enhance the overall performance of the model.

D. FGSI Strategy

1) Structural Components: Remote sensing images often have high resolution, capturing extensive scenes where objects appear as minute details. The tiny objects and dense arrangements make feature extraction challenging for localization tasks, as standard CLIP methods may overlook the finer details of these objects. To address this issue, we utilize the vision transformer (ViT) model pretrained with semantic segmentation task from [21] to initialize our image encoder. This choice allows for extracting image features with multiscale information and detail-rich characteristics essential for smart objects localization in remote sensing.

Most recent visual language models in remote sensing utilize ViT pretrained with the CLIP framework. This type of encoder aligns image features with text representations, enabling LLMs to understand image features. However, for remote sensing scene images, using CLIP-trained models has limitations. CLIP models are primarily trained on internet-based image–text pairs, with relatively little remote sensing data, requiring a substantial volume of remote sensing data for effective transfer learning. Moreover, CLIP's training methodology focuses on global semantic information, potentially overlooking local details critical for interpreting densely clustered objects in remote sensing imagery. These factors underscore the challenges of adapting CLIP-trained models as visual encoders for remote sensing tasks.

In contrast, we initialized our image encoder with SAM, which excels in extracting image features for zero-shot segmentation tasks, emphasizing local details and edge information crucial for localization tasks. The fine-grained image

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encoder has not encountered textual features during the pretraining phase. However, we found that through the VLSP layer and full-parameter training, it is sufficient to get semantic information from the visual features for the language-guided object localization task.

The *VLSP layer*, a simple single-layer linear mapping, contributes to the integration of visual features with language. This layer maps the fine-grained semantic features extracted by the encoder to the dimensions of the language model. Through a straightforward linear transformation, the VLSP layer facilitates the alignment of visual and textual information, ensuring that semantic details are preserved for accurate localization in remote sensing tasks.

2) Training Methodology: We adopt a two-stage, full-parameter training strategy to preserve fine-grained semantic features. Previous methods that freeze the language model while fine-tuning the visual encoder often force the encoder's features to align with the semantic space of the LLM. This alignment can be problematic for remote sensing images, as their complex backgrounds and dense object configurations are difficult to describe adequately using natural language. This misalignment can result in the loss of detailed information captured by the visual encoder's features. In contrast, our full-parameter training method allows for direct supervision of both the visual encoder and the text decoder, ensuring they work together more effectively. This approach maintains the detailed, fine-grained features of the visual encoder while enabling the text decoder to accurately interpret these features for text-based localization tasks. By jointly optimizing both components, we enhance the model's ability to handle the intricacies of remote sensing imagery and improve its performance in language-guided object localization.

Our training process is divided into two stages.

a) Stage 1: In the first stage, we utilize extensive datasets for visual grounding and object detection in natural scenes to stimulate the model's capability in accurately locating objects within images. This initial stage focuses on enabling the model to comprehend the fundamental spatial relationships between objects and adapt to handle BBox coordinates as pure text. The datasets included Object365 [42], COCO2017 [43], and CrowdHuman [44], which focus on identifying and locating objects within images. The visual localization datasets comprise RefCOCO, RefCOCOg, and RefCOCO+ [45], which are specialized in identifying and locating specific object in images based on natural language descriptions.

b) Stage 2: In the second stage, we transfer the knowledge acquired in Stage 1 to the domain of remote sensing. We utilize the RSVG-DIOR [5] dataset and its augmented version for training in remote sensing expression-guided localization tasks. The augmented dataset provides additional variability, which helps improve the model's robustness and generalization capabilities. For remote sensing class-guided localization tasks, we leverage the DIOR [3] dataset. It simulates and enhances the model's ability for dense objects localization, enabling it to effectively identify and process closely spaced objects within complex scenes.

The transition from natural scenes to remote sensing images involves significant domain adaptation. Remote sensing images often present challenges such as higher object density, varying scales, and different spectral characteristics. Through full-parameter training in Stage 2, our model adapts to these challenges, transferring the object localization abilities learned from natural scenes to remote sensing scenarios. This two-stage, full-parameter training strategy ensures that the model not only retains its initial capabilities but also becomes proficient in handling the complexities of remote sensing data, thereby improving its performance in identifying and localizing objects with high precision.

IV. Experiments

In this section, we conducted experiments using the LM-RSE for the remote sensing language-guided object localization task to validate its effectiveness. In Section IV-A, we detail the experimental setup. Section IV-B provide the main results of our method on expression-guided localization and compare them against other methods. Section IV-C demonstrates the potential of our approach in class-guided localization within dense scenes. Finally, an ablation study is conducted in Section IV-D.

A. Implementation Details

Our experiments are implemented on two NVIDIA A800 GPUs. All images are uniformly scaled to a resolution of 1024×1024 pixels to preserve fine-grained details essential for detecting small and densely packed objects in remote sensing imagery. The training process is configured with a batch size of 16 per device, which can be reduced according to the hardware capabilities. We initiate the learning rate at $5e^{-5}$ across all datasets, and it is governed by a cosine learning rate scheduler to optimize performance over the course of training. To accommodate the extensive context of our inputs, we set the token's maximum length to 4096. The image encoder component of the model is initialized with pretrained weights from the SAM [21], while the language decoder is initialized using the weights from the OPT-125M [13] model. The final dimension of visual feature is 1024 and the hidden state of the OPT-125M is 768. The hyperparameter d_{num} was set to 256 as it represents the optimal trade-off between accuracy and model complexity, ensuring a balanced performance for both expression-guided and class-guided object localization tasks. The training procedure is divided into two distinct stages. In the first stage, we utilized datasets focused on object-level interpretation tasks within natural scenes for three epochs. The second stage involves adding the BBR branch and training on remote sensing datasets for ten epochs. During the training process, none of the layers are frozen, allowing for full-parameter training.

B. Expression-Guided Object Localization Results

1) Datasets for Evaluation: RSVGD [5] is a comprehensive visual grounding dataset tailored for remote sensing, built upon the DIOR object detection dataset. It comprises 38 320 language expressions, each uniquely associated with one object BBox across 17 402 remote sensing images. These expressions describe attributes such as color, size, and position for objects

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TABLE I
COMPARISON WITH THE STATE-OF-THE-ART METHODS FOR EXPRESSION-GUIDED LOCALIZATION TASK ON THE TEST SET OF RSVGD

Methods	Pr@0.5(%)	Pr@0.6(%)	Pr@0.7(%)	Pr@0.8(%)	Pr@0.9(%)	meanIoU(%)	cumIoU(%)
LBYL-Net [30]	70.63	65.53	59.89	43.89	15.97	62.13	73.50
TransVG [7]	69.30	64.28	56.82	45.11	22.95	60.45	72.88
VLTVG [36]	74.67	70.98	65.34	55.23	34.52	67.11	78.22
MGVLF [5]	74.28	69.93	62.62	51.56	30.68	65.39	77.15
LPVA [46]	82.28	77.35	72.55	60.98	39.98	72.54	85.41
LQVG [47]	83.53	81.73	75.91	65.02	43.23	74.61	82.41
LM-RSE(Ours)	**84.97**	**83.93**	**81.6**	**76.23**	**62.95**	**78.91**	**87.39**

in 20 different categories. The images share a uniform size of 800 × 800 pixels, with spatial resolutions ranging from 0.5 to 30 m. We follow the dataset split outlined in [5], with 40% of the expressions and their corresponding BBoxs allocated to the training set, 10% to the validation set, and the remaining 50% to the test set. As described in Section III-D2, we employ data augmentation to triple the size of the training set during training stage 2, thereby achieving a balance across various tasks. For evaluation, we use the test sets to measure our model's performance in expression-guided localization.

2) Evaluation Metrics: In the context of evaluating a visual grounding model, the effectiveness of the model in locating and identifying objects from an image based on a query is quantified using several key metrics. Specifically, the performance is assessed using precision at varying IoU thresholds: 0.5, 0.6, 0.7, 0.8, and 0.9, denoted as Pr@0.5, Pr@0.6, Pr@0.7, Pr@0.8, and Pr@0.9, respectively. These metrics indicate the proportion of predictions where the overlap between the predicted and ground-truth BBoxs exceeds the respective IoU thresholds.

Additionally, the mean IoU (meanIoU) and cumulative IoU (cumIoU) are utilized as comprehensive indicators of the model's accuracy. The meanIoU is calculated using the formula

$$\text{meanIoU} = \frac{1}{M}\sum_{t=1}^{M}\frac{I_t}{U_t} \tag{15}$$

where M is the total number of image–query pairs, and I_t and U_t represent the intersection and union areas of the predicted and ground-truth BBoxs for each pair, respectively. The cumIoU, on the other hand, is determined by

$$\text{cumIoU} = \frac{\sum_{t=1}^{M} I_t}{\sum_{t=1}^{M} U_t}. \tag{16}$$

3) Results: As shown in Table I, we present the results of our proposed method, LM-RSE, and compare its performance with state-of-the-art methods on the RSVGD test set. The reported results of the comparison methods are reproduced using the official open-source codes, with all models applying consistent training procedures on the same platform.

LM-RSE achieves the highest precision at all thresholds, especially at high thresholds. At a precision threshold of 0.5, our model achieves a precision of 84.97%, which is 1.56% higher than the SOTA model, LQVG [47], which stands at 83.41%. Similarly, at a precision threshold of 0.6, our model reaches 83.93%, surpassing LQVG by 2.9%. When the precision threshold is set to 0.7 and 0.8, LM-RSE maintains its lead with a precision of 81.6% and 76.23%, respectively, outstripping previous SOTA by 5.69% and 10.71%. Even at the highest precision threshold of 0.9, LM-RSE achieves 62.95%, which is significantly higher than SOTA at 43.53%.

In addition to precision, our model excels in the meanIoU metric, recording a value of 78.91%, which is 4.89% higher than the closest competitor. Moreover, the cumulative IoU for LM-RSE is 87.39%, leading LPVA [46] by 2.28%. It can also be observed that, on all metrics, the tiny version of LM-RSE does not lead to a significant performance drop, despite reducing the input image size. These results indicate that our method is highly effective in accurately localizing objects based on expression in remote sensing images.

These results are reinforced by the qualitative analysis presented in Fig. 6. Rows I-IV display complex scenarios involving the localization of different categories, positions, and sizes. On the left (Column A), we present our results along with zoomed-in views, while on the right (Column B), the failure cases from MGVLF [5] are shown. In row I, the model needs to recognize the color and size of the reference object. Although I-B identifies the object, the BBox is larger than necessary and fails to reflect the keyword "smaller" in the expression. For row II, the expression requires finding an airplane little bigger than the one on the left. Our model correctly locates the slightly larger plane, while II-B ignores the keyword "little" and locates the largest plane. In row III, the "expressway service area" blends with the background, but LM-RSE still accurately locates the service area in the lower right corner. III-B also identifies a service area in the lower right corner, but our model's localization is more to the right, better matching the expression. The expression in row IV instructs the model to find a ship of the same size as the one on the top. This is challenging since the object is small and the size difference is subtle. However, LM-RSE accurately locates the indicated ship, while IV-B identifies a ship that is slightly larger than the one above, failing to capture the size relationship between the objects. The visualized results demonstrate LM-RSE's high precision in localizing and identifying various objects in remote sensing images.

In conclusion, LM-RSE not only achieves superior performance across all quantitative metrics but also demonstrates exceptional practical applicability through accurate and reliable visual predictions, establishing its robustness and effectiveness for expression-guided localization tasks in remote sensing images.

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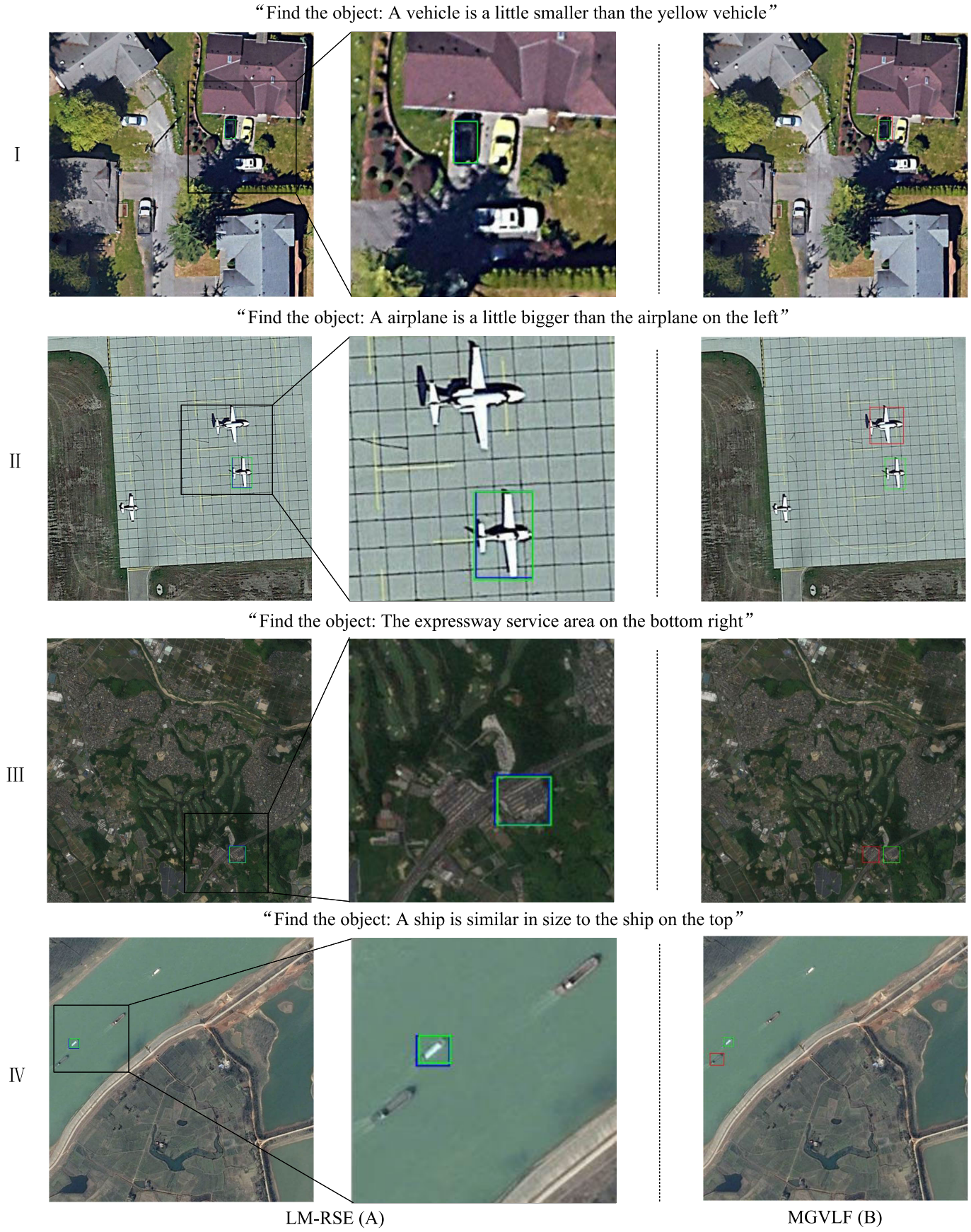


Fig. 6. Comparison of LM-RSE results with MGVLF on RSVG-DIOR dataset. Green boxes indicate ground truth, blue boxes show LM-RSE results, and red boxes highlight errors from MGVLF. Rows I–IV showcase complex scenarios involving the localization of different categories, positions, and sizes.

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TABLE II
CLASS-GUIDED LOCALIZATION PERFORMANCE COMPARISON WITH OTHERS ON UCAS-AOD DATASETS

Methods	Language-guided object localization	Car		Plane		Overall	
		Precision(%)	Recall(%)	Precision(%)	Recall(%)	Precision(%)	Recall(%)
YOLO_RS	✗	74.61	72.50	89.52	90.21	81.88	81.13
GeoChat	✓	21.80	20.24	35.82	39.43	28.63	29.59
LM-RSE	✓	56.44	43.13	79.79	65.52	67.82	54.04

C. Class-Guided Object Localization Results

1) Datasets for Evaluation: The UCAS-AOD dataset [48] is an aerial object detection dataset that includes images of cars and planes collected from Google Earth aerial images. It consists of 510 images with 7114 car samples and 1000 images with 7482 plane samples, all with various object orientations. The images have a resolution of 1280 × 659 pixels. We have reformatted the entire dataset into a class-guided dialog format, as described in Section III-B to evaluate our model's performance. The dataset was randomly split, with 80% used for training and 20% for testing.

2) Evaluation Metrics: We utilize precision and recall to assess the performance of class-guided localization task, which outputs BBoxs for all objects of the same category in an image. We consider an object to be correctly localized if its IoU with the ground truth exceeds 0.5. Specifically, precision and recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

where TP represents true positives, meaning the number of correctly localized objects, where the IoU exceeds 0.5. FP represents false positives, occurring when a BBox is predicted by the model but does not contain an object (or contains an incorrect object). FN represents false negatives, meaning the number of the model fails to predict a BBox for an object that is present in the ground truth.

These metrics are computed for each category, and the overall precision and recall for all categories are also computed during evaluation.

3) Results: Table II presents a comparison between our proposed method, LM-RSE, with the traditional object detection model YOLO_RS and the remote sensing multimodal model GeoChat [10]. YOLO_RS is a model fine-tuned from YOLOv3 [49] on remote sensing datasets. This comparison focuses on class-guided localization tasks using the UCAS-AOD dataset, specifically for two categories: cars and planes. The YOLOv3 model, as a specialized object detection model, performs localization and classification tasks for all objects in the image, but it cannot perform class-specific localization guided by language. Therefore, we use it as a benchmark model to assess the localization capability of vision-language models. In comparison with GeoChat, LM-RSE shows competitive results across the board. For the car category, our method achieves a precision of 56.4% and a recall of 42.5%, significantly outperforming GeoChat, which only manages a precision of only 21.8% and a recall of 20.2%. Similarly, in the plane category, LM-RSE records a precision of 79.2% and a recall of 65.5%, whereas GeoChat lags far behind with a precision of 35.8% and a recall of 39.4%.

The metrics indicate that the precision and recall for the car category are generally lower compared to those for the plane category. This is due to the fact that car objects in remote sensing images are typically smaller and more densely packed, making localization more challenging. Enhancing the language model's ability to accurately localize dense, small objects will be a focus of future research. The overall average metrics, with LM-RSE achieving a precision of 67.5% and a recall of 54.0%, show that its category-guided multiobject localization capability significantly surpasses that of GeoChat, and there is still room for improvement compared to specialized object detection models.

The visualized results in Fig. 7 provide additional insights. The first two rows compare the class-guided object localization results of our model and GeoChat. The first row shows the results of GeoChat, with red boxes representing GeoChat's predicted BBoxs and green boxes indicating the ground truth. It is evident that the model fails to accurately localize most of the objects. The second row shows the results from our model. The blue bboxes represent our predicted bounding bboxes, which in many cases fit the edges of the objects even more accurately than the ground truth, marked by green bboxes. For the plane category, our predictions closely align with the ground truth. While in the car category, although some smaller cars were missed, our model localized an unannotated truck in the center of the second image. While our model demonstrates strong localization capabilities, it faces challenges in certain complex lighting conditions, as illustrated in the third row of Fig. 7. These limitations highlight areas for future improvement, particularly in distinguishing shadows from dark objects and in handling objects with similar appearances. Nevertheless, these challenges provide valuable insights for the continued development of more robust models.

In conclusion, the results from both the table and the visualizations illustrate that LM-RSE significantly outperforms the GeoChat model in the class-guided localization task. The model not only achieves higher precision and recall but also demonstrates exceptional visual accuracy in identifying and localizing based on the class name.

D. Ablation Study

In order to thoroughly evaluate the contribution of our proposed method, we conducted a series of ablation experiments. We tested the effectiveness of the BBR approach and the FGSI strategy's components and explored the optimal combinations. This section details the design and results of these ablation studies, highlighting the significance of each component in achieving superior performance.

1) BBR Approach: As shown in Table III, we analyzed the effectiveness of the box decoder and the two parts of the regression loss. The first row presents the localization results without the BBR approach, which leads to a decrease of 2.24% in Pr@0.5 for the expression-guided object localization task and a 5.4% reduction in precision for the class-guided object localization task. In the second row, we incorporate the box decoder and use only the center loss function, which improves

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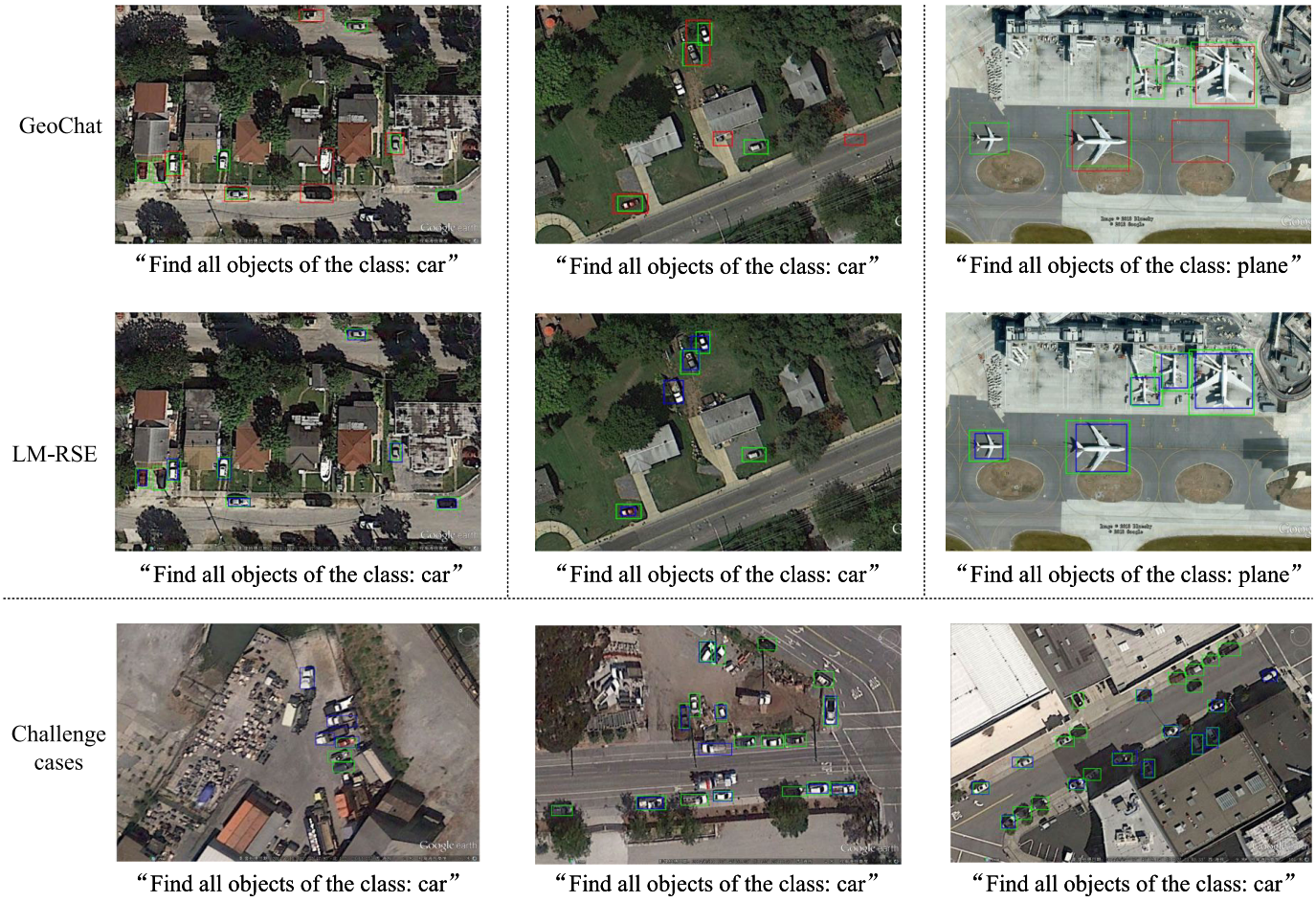


Fig. 7. Visualization of class-guided object localization results on the UCAS-AOD dataset. The first two rows compare the performance of LM-RSE and GeoChat. Ground-truth BBoxs are shown in green, LM-RSE results in blue, and GeoChat results in red. The third row illustrates challenging cases where our model fails to perform accurately.

TABLE III
ABLATION STUDIES OF THE BBR APPROACH

Box Decoder	Regression Loss		Pr@0.5 (Expression-guided)	Precision (Class-guided)
	L_{center}	L_{bbox}		
✗	✗	✗	82.73	62.12
✓	✓	✗	84.01(↑1.28)	64.23(↑2.11)
✓	✗	✓	84.38(↑1.65)	64.61(↑2.49)
✓	✓	✓	84.97(↑2.24)	67.82(↑5.70)

TABLE IV
ABLATION STUDIES FOR DIFFERENT MLP DEPTHS AND TOKEN COUNTS IN THE BBR APPROACH

BBox Decoder Depth Comparison		<BBox> Count Comparison	
Depth	meanIOU (%)	Count	meanIOU (%)
2	74.15	1	78.83
3	78.91	2	78.91
4	75.18	4	76.91

performance, yielding a Pr@0.5 of 84.01% and a precision of 64.6%, demonstrating an increase over the baseline model. The third row displays the results of our proposed model, which includes all components. This model achieves the highest Pr@0.5 and precision on both tasks, further validating the effectiveness of the BBR approach.

Table IV examines the impact of different BBox decoder depths and the number of special <BBox> tokens on model performance. We evaluated decoder depths of two, three, and four layers, and <BBox> token counts of 1, 2, and 4, respectively. During each test, the other components were kept at their optimal settings. The results indicate that a three-layer BBox decoder with 2 <BBox> tokens yields the best performance, achieving a mean IOU of 78.91%. Increasing or decreasing either component leads to a drop in mean IOU. This suggests that this configuration strikes an optimal balance between model complexity and generalization ability on the RSVGD dataset.

2) FGSI Strategy: As shown in Table V, we evaluated the impact of different components in the FGSI strategy through ablation studies. Model (a), which uses the CLIP pre-trained ViT as the image encoder instead of our fine-grained encoder, fails to extract fine-grained visual features, resulting in a significant drop of 10.06% in meanIoU and 7.25% in Pr@0.5 for expression-guided localization. Model (b) replaces our VLSP layer with a more complex three-layer MLP, aimed at exploring the impact of a more sophisticated mapping structure. While Model (b) achieves slight improvements in certain

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TABLE V
ABLATION STUDIES FOR FGSI STRATEGY IN OUR NETWORK

Model	Fine-grained Encoder	VLSP Layer	Full-parameter	Two-stage	Pr@0.5(%)	meanIOU (%)
(a)	✗	✓	✓	✓	77.72%	68.85%
(b)	✓	✗	✓	✓	84.96%	78.92%
(c)	✓	✓	✗	✓	68.39%	59.95%
(d)	✓	✓	✓	✗	79.72%	72.85%
Ours	✓	✓	✓	✓	84.97%	78.91%

TABLE VI
ABLATION STUDY ON THE INPUT IMAGE SIZE

Input Size	Expression-guided		Class-guided	
	Pr@0.5(%)	meanIoU(%)	Precision(%)	Recall(%)
512×512	81.06	74.59	50.85	41.15
800×800	83.73	77.32	61.92	48.65
1024×1024	**84.97**	**78.91**	**67.82**	**54.04**

metrics, such as meanIoU, these gains are marginal and come at the cost of significantly increased model complexity and computational overhead. This indicates that the VLSP layer, despite its simplicity, provides competitive performance with much lower computational cost. On the other hand, Model (c), which ablates the full-parameter training strategy and only fine-tunes the language decoder, suffers a severe accuracy loss, yielding a Pr@0.5 of 68.39% and a meanIoU of 59.95%. Finally, Model (d), which omits the two-stage training process and applies only the Stage 2 training, shows a performance drop of approximately 6.06% in meanIoU and 5.25% in Pr@0.5 without the benefit of Stage 1.

Table VI shows the impact of different input image resolutions on model performance. As the input resolution directly determines the number of visual tokens generated by the image encoder, smaller resized images result in reduced representational capacity and lower accuracy. For example, resizing inputs to 512 × 512 leads to a noticeable drop in performance, with Pr@0.5 of 81.06% and precision of only 50.85%. Although the 800 × 800 resolution—matching the original size of the expression-guided dataset—achieves competitive results in that task, it struggles to model small and densely distributed objects in the class-guided task, yielding a precision of 61.92%. In contrast, using 1024 × 1024 consistently yields the best performance across both tasks, demonstrating its effectiveness in preserving spatial detail and improving localization accuracy.

These results underscore the importance of each component in the FGSI strategy, confirming that the fine-grained encoder, VLSP layer, full-parameter training, and two-stage training process are all critical to achieving optimal performance.

V. CONCLUSION

In this article, we propose LM-RSE, a lightweight generative model for remote sensing language-guided object localization named LM-RSE. It uses a generative way to handle both expression-guided and class-guided localization, and uses a regressive approach to increase the precision and accuracy of the model. More specifically, we introduce a novel BBR approach to enhance numerical prediction accuracy and a fine-grained semantic integration (FGSI) strategy to preserve semantic integrity of the remote sensing image. LM-RSE achieves state-of-the-art performance on the RSVGD dataset for expression-guided localization and excels in high-precision metrics. The results of class-guided localization also demonstrate the model's potential to tackle dense scenes, paving the way for more advanced and flexible applications in remote sensing.

Future work: While our LM-RSE shows satisfactory performance in class-guided localization tasks, there is room for improvement. Future efforts will focus on enhancing the model's capabilities in this area, aiming to achieve more precise and reliable results across various object categories.

In addition to improving the class-guided localization's performance, we plan to explore new directions in language-guided object localization tasks. Specifically, we will investigate the application of our model to rotated BBoxs, which are particularly relevant for remote sensing scenarios where objects are often not aligned with the image axes. Furthermore, we are working on extending our research to language-guided object localization in video sequences, which presents unique challenges such as temporal consistency, dynamic object interactions, and more abundant language descriptions.

These extensions of our work are already underway, and we anticipate that they will open new avenues for improving the robustness and versatility of language-guided object localization models in diverse and complex environments.

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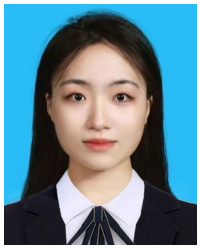
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